On the Suboptimality of Rate–Distortion-Optimal Compression: Fundamental Accuracy Limits for Distributed Localization

Amir Weiss, *Senior Member, IEEE*

***Abstract*—We derive fundamental accuracy limits for distributed localization when a fusion center has access only to independently rate–distortion (RD)-optimally compressed versions of multi-sensor observations, under a line-of-sight propagation model with a Gaussian wideband waveform. Using the Gaussian RD test-channel model together with a Whittle spectral Fisher-information characterization, we obtain an explicit frequency-domain Cramér–Rao lower bound. A two-band, two-level specialization yields closed-form expressions and reveals a rate-induced regime change: RD-optimal compression under a squared-error distortion measure can eliminate localization-informative spectral content. A simple band-selective scheme can outperform RD compression by orders of magnitude at the same rate, motivating localization-aware compression for networked sensing and integrated sensing and communication systems.**

***Index Terms*—Distributed localization, rate–distortion theory, task-oriented compression, Cramér–Rao lower bound.**

I. Introduction

THE proliferation of distributed sensing and communication platforms—including Internet-of-Things deployments [1], sensor networks [2], wearable devices [3], and edge-assisted cyber–physical systems [4]—has accelerated the need to operate under stringent resource constraints. In such settings, raw waveform streaming from multiple nodes to a fusion center (FC) is often infeasible due to limited uplink rates, energy budgets, latency requirements, or shared-spectrum constraints. As a result, modern architectures increasingly rely on *compressed observations* [5], where each node transmits a rate-constrained representation of its measurements [6], while inference is performed centrally or cooperatively. Understanding the fundamental impact of compression on estimation fidelity [7] is therefore critical for principled system design.

A particularly important inference task is *distributed localization*, which underpins numerous applications such as navigation, robotics, and industrial monitoring [8]. It has also emerged as a key capability in integrated sensing and communications (ISAC) [9], [10], [11], where distributed receivers collect observations that must often be relayed over rate-limited links. This makes the interplay between compression and localization accuracy a first-order design consideration.

Despite extensive work on localization and on compression for estimation in relative isolation (e.g., [12], [13], [14], [15], [16]), including task-oriented quantization perspectives [6], a Cramér–Rao lower bound (CRLB) characterization for time-delay estimation (TDE) in dispersed-spectrum settings [17], and CRLB–versus–rate tradeoffs for rate-constrained distributed TDE from quantized observations [18], the literature that provides *analytically explicit* accuracy limits for localization *from compressed multi-sensor observations* remains relatively sparse, especially in wideband waveform models where time-of-arrival structure is central. In particular, while rate–distortion (RD) theory [19] offers a tractable lens for modeling optimal compression under a squared-error distortion criterion, it is unclear how RD-optimal compression for *signal reconstruction* translates into localization performance, and whether compress-then-estimate design is well aligned with the information content most relevant for localization.

This letter addresses the following fundamental question: *What are the accuracy limits of distributed localization when the FC has access only to independently compressed versions of the sensor observations?* Under a line-of-sight wideband Gaussian waveform model, we derive frequency-domain CRLBs for localization from compressed observations by combining a spectral Fisher-information (FI) characterization with a Gaussian test-channel model of compression. We specialize the bound to a two-band, two-level spectral model, yielding closed-form expressions that reveal nontrivial rate-dependent behavior, including regime changes induced by water-filling. Finally, we construct an explicit counterexample showing that compression optimized for mean-square signal reconstruction can be strictly suboptimal for localization accuracy, which motivates localization-aware (task-oriented) compression beyond compress-then-estimate designs.

II. Problem Formulation

We consider a distributed sensing system with $M \geq 2$ spatially separated sensors. Let $\mathbf{p}_m \in \mathbb{R}^{d\times 1}$ denote the known position of sensor $m \in \{1,\ldots,M\}$, and let $\mathbf{p} \in \mathbb{R}^{d\times 1}$ denote the unknown source location, where typically $d \in \{2,3\}$. Assuming line-of-sight propagation with known speed $c > 0$, the propagation delay to sensor m is

$$\tau_m(\mathbf{p}) \triangleq \frac{\|\mathbf{p}-\mathbf{p}_m\|_2}{c} \in \mathbb{R}_+, \quad m \in \{1,\ldots,M\}. \tag{1}$$

As will be evident from the second-order statistics below, and as is standard in passive localization, the dependence on $\mathbf{p}$ is only

Received 30 January 2026; revised 22 March 2026; accepted 23 March 2026. Date of publication 26 March 2026; date of current version 28 April 2026. The associate editor coordinating the review of this article and approving it for publication was Dr. Lu Gan.

The author is with The Alexander Kofkin Faculty of Engineering, Bar-Ilan University, Ramat Gan 5290002, Israel (e-mail: amir.weiss@biu.ac.il).

This article has supplementary downloadable material available at https://doi.org/10.1109/LSP.2026.3678117, provided by the author.

Digital Object Identifier 10.1109/LSP.2026.3678117

1070-9908 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

through the time-delay differences $\Delta_{m\ell}(\mathbf{p}) \triangleq \tau_m(\mathbf{p}) - \tau_\ell(\mathbf{p})$. Thus, no "global time-shift" parameter is introduced.

Each sensor observes a noisy, time-shifted version of a common wideband source over the interval $t \in [0, T]$,

$$\mathsf{x}_m(t) = \mathsf{s}\left(t - \tau_m(\mathbf{p})\right) + \mathsf{n}_m(t) \in \mathbb{R}, \quad m \in \{1, \ldots, M\}, \tag{2}$$

where $\mathsf{s}(t)$ is a zero-mean wide-sense stationary (WSS) Gaussian process with (two-sided) power spectral density (PSD) function $S_{\mathsf{s}}(f)$ supported on $|f| \le B$ for some known $B > 0$. The noises $\{\mathsf{n}_m(t)\}_{m=1}^M$ are mutually independent, independent of $\mathsf{s}(t)$, zero-mean WSS Gaussian processes with PSDs $\{S_{\mathsf{n}_m}(f)\}_{m=1}^M$ (supported on $|f| \le B$).

Define the sensor vector process $\mathbf{x}(t) \triangleq [\mathsf{x}_1(t) \ \cdots \ \mathsf{x}_M(t)]^{\mathsf{T}}$. Its (matrix-valued) cross-spectral density admits the form

$$\mathbf{S}_{\mathbf{x}}(f; \mathbf{p}) = S_{\mathsf{s}}(f)\, \mathbf{v}(f; \mathbf{p}) \mathbf{v}^{\mathsf{H}}(f; \mathbf{p}) + \mathbf{S}_{\mathbf{n}}(f) \in \mathbb{C}^{M \times M}, \tag{3}$$

where $\mathbf{S}_{\mathbf{n}}(f) \triangleq \operatorname{diag}(S_{\mathsf{n}_1}(f), \ldots, S_{\mathsf{n}_M}(f))$ and

$$\mathbf{v}(f; \mathbf{p}) \triangleq \left[e^{-\jmath 2\pi f \tau_1(\mathbf{p})} \ \cdots \ e^{-\jmath 2\pi f \tau_M(\mathbf{p})}\right]^{\mathsf{T}} \in \mathbb{C}^{M \times 1}. \tag{4}$$

In particular, for $m \neq \ell$,

$$[\mathbf{S}_{\mathbf{x}}(f; \mathbf{p})]_{m\ell} = S_{\mathsf{s}}(f)\, e^{-\jmath 2\pi f \Delta_{m\ell}(\mathbf{p})} \in \mathbb{C}, \tag{5}$$

which shows explicitly that $\mathbf{p}$ affects the observation law only through time-difference-of-arrival terms.

In the distributed setting with communication constraints, each sensor m communicates to a FC over a rate-limited link of rate R_m bits per second. Thus, the FC does not have direct access to $\mathsf{x}_m(\cdot)$, but rather to a compressed version thereof, denoted as $\widehat{\mathsf{x}}_m(\cdot)$, produced at sensor m under the rate constraint R_m. Let $\widehat{\mathbf{x}}(t) \triangleq [\widehat{\mathsf{x}}_1(t) \ \cdots \ \widehat{\mathsf{x}}_M(t)]^{\mathsf{T}}$ denote the vector process of the compressed signals available at the FC.

Based on $\widehat{\mathbf{x}}(\cdot)$ over $[0, T]$, the FC constructs an estimator $\widehat{\mathbf{p}}$ of the unknown location $\mathbf{p}$. The focus of this letter is to characterize the fundamental localization limits from RD-optimally compressed observations by deriving the Cramér–Rao lower bound (CRLB) on the mean-square error of any unbiased estimator of $\widehat{\mathbf{p}}$ based on $\widehat{\mathbf{x}}(\cdot)$ over $[0, T]$. In addition, and perhaps surprisingly, we will show that it is easy to design a strictly RD-suboptimal compression giving significantly higher localization accuracy, thus highlighting the need for developing joint compression-localization schemes.

III. CRLB for RD-Optimally Compressed Signals

Under the Gaussian signal model in Section II and the squared-error distortion measure, the RD optimal compression of a WSS Gaussian process admits a convenient test-channel representation (e.g., [19]). In particular, the m-th compressed waveform available at the FC can be modeled as

$$\widehat{\mathsf{x}}_m(t) = (b_m * \mathsf{x}_m)(t) + \mathsf{z}_m(t) \in \mathbb{R}, \tag{6}$$

where $b_m(t)$ is a linear time-invariant (LTI) filter with (real-valued) frequency response $B_m(f)$, and $\mathsf{z}_m(t)$ is a zero-mean WSS Gaussian "compression-noise" process, independent of $\mathsf{x}_m(t)$, with PSD $S_{\mathsf{z}_m}(f)$. Denoting by $S_{\mathsf{x}_m}(f)$ the PSD of $\mathsf{x}_m(t)$, the RD water-filling solution [20] implies that there exists a water level $\lambda_m \ge 0$ such that

$$B_m(f) = \left[1 - \frac{\lambda_m}{S_{\mathsf{x}_m}(f)}\right]^+, \quad S_{\mathsf{z}_m}(f) = \lambda_m\, B_m(f), \tag{7}$$

where $[u]^+ \triangleq \max\{u, 0\}$. The water level λ_m is uniquely determined by the rate constraint R_m via

$$R_m = \frac{1}{2} \int_{-\infty}^{\infty} \left[\log_2\left(\frac{S_{\mathsf{x}_m}(f)}{\lambda_m}\right)\right]^+ \mathrm{d}f, \tag{8}$$

where the specialized integral (8) in our bandlimited signal setting is effectively over $|f| \le B$.

Since each $\widehat{\mathsf{x}}_m(t)$ in (6) is obtained from a linear transformation of $\mathsf{x}_m(t)$ with an addition of independent WSS Gaussian noise, the compressed vector process $\widehat{\mathbf{x}}(t)$ remains WSS Gaussian. Consequently, its statistical law is fully characterized by its cross-spectral density matrix $\mathbf{S}_{\widehat{\mathbf{x}}}(f; \mathbf{p})$, given by

$$\mathbf{S}_{\widehat{\mathbf{x}}}(f; \mathbf{p}) = S_{\mathsf{s}}(f)\, \mathbf{B}(f) \mathbf{v}(f; \mathbf{p}) \mathbf{v}^{\mathsf{H}}(f; \mathbf{p}) \mathbf{B}(f) + \mathbf{\Sigma}(f), \tag{9}$$

where we have defined

$$\mathbf{B}(f) \triangleq \operatorname{diag}(B_1(f), \ldots, B_M(f)) \in \mathbb{R}^{M \times M}, \tag{10}$$

and using $\{S_{\mathsf{w}_m}(f) \triangleq |B_m(f)|^2 S_{\mathsf{n}_m}(f) + S_{\mathsf{z}_m}(f)\}_{m=1}^M$,

$$\mathbf{\Sigma}(f) \triangleq \operatorname{diag}(S_{\mathsf{w}_1}(f), \ldots, S_{\mathsf{w}_M}(f)) \in \mathbb{R}^{M \times M}. \tag{11}$$

A. CRLB via a Spectral FI

Since $\widehat{\mathbf{x}}(t)$ is WSS Gaussian with a cross-spectral density matrix $\mathbf{S}_{\widehat{\mathbf{x}}}(f; \mathbf{p})$, the localization information can be expressed in the frequency domain. Specifically, under standard regularity conditions for purely non-deterministic stationary Gaussian processes,[1] the FI rate (i.e., per unit time) matrix admits the Whittle spectral representation [21], i.e.,

$$\lim_{T \to \infty} \frac{1}{T} \left[\mathbf{J}^{\mathsf{RD}}(\mathbf{p})\right]_{ij} \tag{12}$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} \operatorname{tr}\left(\mathbf{S}_{\widehat{\mathbf{x}}}^{-1}(f; \mathbf{p}) \mathbf{S}'_{\widehat{\mathbf{x}},i}(f; \mathbf{p}) \mathbf{S}_{\widehat{\mathbf{x}}}^{-1}(f; \mathbf{p}) \mathbf{S}'_{\widehat{\mathbf{x}},j}(f; \mathbf{p})\right) \mathrm{d}f, \tag{13}$$

where, for all $1 \le i, j \le d$, $[\mathbf{J}^{\mathsf{RD}}(\mathbf{p})]_{ij}$ denotes the (i, j)-th element of the FI rate matrix for the RD-optimally compressed signals $\widehat{\mathbf{x}}(\cdot)$, $\mathbf{S}'_{\widehat{\mathbf{x}},i}(f; \mathbf{p}) \triangleq \frac{\partial \mathbf{S}_{\widehat{\mathbf{x}}}(f; \mathbf{p})}{\partial p_i}$, and in our bandlimited model the integral reduces to $f \in [-B, B]$. For an observation horizon T, the CRLB is (e.g., [22])

$$\mathbb{C}\mathrm{ov}\left(\widehat{\mathbf{p}} - \mathbf{p}\right) \triangleq \mathbb{E}\left[\left(\widehat{\mathbf{p}} - \mathbf{p}\right)\left(\widehat{\mathbf{p}} - \mathbf{p}\right)\right] \succeq \mathbf{J}_T^{\mathsf{RD}\,-1}(\mathbf{p}), \tag{14}$$

where $\mathbf{J}_T^{\mathsf{RD}}(\mathbf{p})$ denotes the FI matrix (FIM) for $\mathbf{p}$ based on $\widehat{\mathbf{x}}(\cdot)$ over $[0, T]$. Moreover, $\mathbf{J}_T^{\mathsf{RD}}(\mathbf{p})$ grows linearly with T, and

$$\mathbf{J}_\infty^{\mathsf{RD}}(\mathbf{p}) \triangleq \lim_{T \to \infty} \frac{1}{T} \mathbf{J}_T^{\mathsf{RD}}(\mathbf{p}) \tag{15}$$

exists and is given by (12). Thus, $\mathbf{J}_T^{\mathsf{RD}}(\mathbf{p}) = T\, \mathbf{J}_\infty^{\mathsf{RD}}(\mathbf{p}) + o(T)$. To evaluate (13), it remains to compute $\mathbf{S}'_{\widehat{\mathbf{x}},i}(f; \mathbf{p})$. Recall (9), and that $\mathbf{\Sigma}(f)$ is independent of $\mathbf{p}$. Thus, the dependence in $\mathbf{p}$ is only through the phase terms $e^{-\jmath 2\pi f \tau_m(\mathbf{p})}$ in $\mathbf{v}(f; \mathbf{p})$. Differentiating (1) yields, for all $1 \le i \le d$,

$$\frac{\partial \tau_m(\mathbf{p})}{\partial p_i} = \frac{1}{c} \frac{p_i - [\mathbf{p}_m]_i}{\|\mathbf{p} - \mathbf{p}_m\|_2}, \quad m \in \{1, \ldots, M\}, \tag{16}$$

and consequently, for each m,

$$\frac{\partial [\mathbf{v}(f; \mathbf{p})]_m}{\partial p_i} = -\jmath 2\pi f \frac{\partial \tau_m(\mathbf{p})}{\partial p_i} [\mathbf{v}(f; \mathbf{p})]_m. \tag{17}$$

Using the notation $\mathbf{v}'_i(f; \mathbf{p}) \triangleq \frac{\partial \mathbf{v}(f; \mathbf{p})}{\partial p_i}$, by the product rule,

$$\mathbf{S}'_{\widehat{\mathbf{x}},i}(f; \mathbf{p}) = S_{\mathsf{s}}(f)\, \mathbf{B}(f) \mathbf{Q}_i(f; \mathbf{p}) \mathbf{B}(f), \tag{18}$$

[1] For example, it is sufficient to assume that $\mathbf{S}_{\widehat{\mathbf{x}}}(f; \mathbf{p})$ is uniformly bounded and positive definite on $[-B, B]$, and is continuously differentiable in $\mathbf{p}$.

where $\mathbf{Q}_i(f;\mathbf{p}) \triangleq \mathbf{v}_i'(f;\mathbf{p})\mathbf{v}^{\mathsf{H}}(f;\mathbf{p}) + \mathbf{v}(f;\mathbf{p})\mathbf{v}_i'^{\mathsf{H}}(f;\mathbf{p})$, and which together with (12) yields an explicit CRLB expression as a one-dimensional integral over $f \in [-B, B]$. We next specialize this expression to a two-band, two-level spectral model to obtain fully closed-form bounds and to show, with a simple RD-suboptimal compression, that RD-optimal compression for reconstruction can be strictly suboptimal for localization.

IV. RD is Not Localization-Optimal: A Two-Band Counterexample

The CRLB expression derived in Section III is explicit, but still involves a frequency integral over the (possibly arbitrary) source PSD $S_{\mathsf{s}}(f)$, and the RD water-filling induces rate-dependent changes in the effective spectrum seen at the FC. It is therefore generally unclear whether RD-optimality for waveform reconstruction transfers to optimality in terms of localization accuracy (or even preserves the most localization-informative spectral components). In this section, we adopt an explicit two-band, two-level spectral model that (i) yields closed-form expressions for the CRLB; and (ii) already captures the key phenomenon: RD-optimal compression for signal reconstruction can be strictly suboptimal for localization.

A. Two-Band, Two-Level Spectral Model

Fix $0 < f_{\mathrm{L}} < f_{\mathrm{H}} \le B$ and define the two disjoint bands

$$\mathcal{B}_{\mathrm{L}} \triangleq \{f : |f| \le f_{\mathrm{L}}\}, \qquad \mathcal{B}_{\mathrm{H}} \triangleq \{f : f_{\mathrm{L}} < |f| \le f_{\mathrm{H}}\}. \tag{19}$$

We assume the following piecewise-constant source spectrum

$$S_{\mathsf{s}}(f) = \begin{cases} S_{\mathrm{L}}, & f \in \mathcal{B}_{\mathrm{L}}, \\ S_{\mathrm{H}}, & f \in \mathcal{B}_{\mathrm{H}}, \\ 0, & |f| > f_{\mathrm{H}}, \end{cases} \tag{20}$$

with $S_{\mathrm{L}} > S_{\mathrm{H}} > 0$. For simplicity, we also assume identical sensor noises with flat PSD in the relevant band,

$$S_{\mathsf{n}_m}(f) = \begin{cases} N_0, & |f| \le f_{\mathrm{H}}, \\ 0, & |f| > f_{\mathrm{H}}, \end{cases} \quad \forall m \in \{1, \dots, M\}, \tag{21}$$

so that $S_{\mathsf{x}_m}(f) = S_{\mathsf{s}}(f) + N_0$ for $|f| \le f_{\mathrm{H}}$.

Under (20)–(21), the RD water-filling solution (7) is also piecewise constant over the two bands. In particular, letting λ_m denote the water level at sensor m, we have

$$B_m(f) = \begin{cases} \left[1 - \frac{\lambda_m}{S_{\mathrm{L}}+N_0}\right]^+, & f \in \mathcal{B}_{\mathrm{L}}, \\ \left[1 - \frac{\lambda_m}{S_{\mathrm{H}}+N_0}\right]^+, & f \in \mathcal{B}_{\mathrm{H}}, \\ 0, & |f| > f_{\mathrm{H}}, \end{cases} \tag{22}$$

and $S_{\mathsf{z}_m}(f) = \lambda_m B_m(f)$ for $|f| \le f_{\mathrm{H}}$. As R_m decreases, λ_m increases and a *regime change* occurs at $\lambda_m = S_{\mathrm{H}} + N_0$:

- **High compression rates:** when $\lambda_m < S_{\mathrm{H}} + N_0$, both bands are active and $B_m(f) > 0$ on $\mathcal{B}_{\mathrm{L}} \cup \mathcal{B}_{\mathrm{H}}$.
- **Intermediate compression rates:** when $S_{\mathrm{H}} + N_0 \le \lambda_m < S_{\mathrm{L}} + N_0$, then $B_m(f) > 0$ on $\mathcal{B}_{\mathrm{L}}$ but $B_m(f) = 0$ on $\mathcal{B}_{\mathrm{H}}$. Critically, the RD solution *drops the high-frequency band*.
- **Low compression rates:** when $\lambda_m \ge S_{\mathrm{L}} + N_0, B_m(f) \equiv 0$ and no information is conveyed.

Thus, in this two-band two-level spectral model, RD-optimal compression can eliminate an entire band as the rate decreases.

However, recall from (17) that the geometry dependence enters through the phase terms $e^{-\jmath 2\pi f \tau_m(\mathbf{p})}$, and differentiation introduces a factor of f. Consequently, the integrand in (13) contains an intrinsic *frequency-squared* weighting: each derivative matrix $\mathbf{S}'_{\widehat{\mathbf{x}},i}(f;\mathbf{p})$ is proportional to f, hence the trace term $\mathrm{tr}(\mathbf{S}_{\widehat{\mathbf{x}}}^{-1}\mathbf{S}'_{\widehat{\mathbf{x}},i}\mathbf{S}_{\widehat{\mathbf{x}}}^{-1}\mathbf{S}'_{\widehat{\mathbf{x}},j})$ scales as f^2 (up to other spectral factors). Intuitively, high-frequency components can therefore be substantially more informative for localization, even when they carry less signal energy.

B. Closed-Form CRLB Under the Two-Band Model

We now specialize (13) under (20)–(22) and derive a closed-form expression. For clarity of exposition (and since the phenomenon is per-sensor), we assume the symmetric setting

$$R_m = R \;\Rightarrow\; \begin{cases} \lambda_m = \lambda, \\ B_m(f) = B(f), \end{cases} \quad \forall m \in \{1, \dots, M\}. \tag{23}$$

For $b \in \{\mathrm{L}, \mathrm{H}\}$, define the per-band effective SNR level

$$\gamma_b(R) \triangleq \frac{S_b\, B_b^2(R)}{S_{\mathsf{w},b}(R)}, \quad S_{\mathsf{w},b}(R) \triangleq B_b^2(R) N_0 + \lambda(R)\, B_b(R), \tag{24}$$

where $S_{\mathrm{L}}, S_{\mathrm{H}}$ are given in (20), $\lambda(R)$ is the RD water level satisfying (8), and $B_b(R) \triangleq B(f)$ for any $f \in \mathcal{B}_b$ (which is constant over each band under (22)). Define the band information weights,

$$w_b(R) \triangleq \frac{2M\gamma_b(R)^2}{1 + M\gamma_b(R)}, \qquad b \in \{\mathrm{L}, \mathrm{H}\}, \tag{25}$$

and with it the lower and higher frequency information terms,

$$J_{\mathrm{L}}(R) \triangleq f_{\mathrm{L}}^3 w_{\mathrm{L}}(R), \quad J_{\mathrm{H}}(R) \triangleq \left(f_{\mathrm{H}}^3 - f_{\mathrm{L}}^3\right) w_{\mathrm{H}}(R). \tag{26}$$

Finally, define (entrywise) the geometry matrix $\mathbf{G}(\mathbf{p}) \in \mathbb{R}^{d\times d}$,

$$[\mathbf{G}(\mathbf{p})]_{ij} \triangleq \widetilde{\mathbf{g}}_i^{\mathsf{T}}(\mathbf{p})\widetilde{\mathbf{g}}_j(\mathbf{p}), \qquad \widetilde{\mathbf{g}}_i(\mathbf{p}) \triangleq \mathbf{P}\mathbf{g}_i(\mathbf{p}), \tag{27}$$

where $\mathbf{P} \triangleq \mathbf{I}_M - \frac{1}{M}\mathbf{1}\mathbf{1}^{\mathsf{T}} \in \mathbb{R}^{M\times M}$ is a projection matrix and

$$\mathbf{g}_i(\mathbf{p}) \triangleq \left[\frac{\partial \tau_1(\mathbf{p})}{\partial p_i} \;\cdots\; \frac{\partial \tau_M(\mathbf{p})}{\partial p_i}\right]^{\mathsf{T}} \in \mathbb{R}^{M\times 1}. \tag{28}$$

We are now ready to state our main result.

Theorem 1 (Closed-form FI rate and CRLB under the two-band model): Under the two-band, two-level spectrum model (20)–(21), RD-optimal compression (22) and the symmetric setting (23), the FI rate for $\mathbf{p}$ based on $\widehat{\mathbf{x}}(\cdot)$ over $[0, T]$ reads

$$\mathbf{J}_{\infty}^{\mathsf{RD}}(\mathbf{p}) = \frac{4\pi^2}{3}\left(J_{\mathrm{L}}(R) + J_{\mathrm{H}}(R)\right)\mathbf{G}(\mathbf{p}). \tag{29}$$

Consequently, for any unbiased estimator $\widehat{\mathbf{p}}$ of $\mathbf{p}$ based on $\widehat{\mathbf{x}}(\cdot)$ over $[0, T]$, it holds that

$$\mathbb{C}\mathrm{ov}(\widehat{\mathbf{p}} - \mathbf{p}) \succeq \mathbf{J}_T^{\mathsf{RD}\,-1}(\mathbf{p}) = \frac{1}{T}\mathbf{J}_{\infty}^{\mathsf{RD}\,-1}(\mathbf{p}) + o\left(\frac{1}{T}\right) \tag{30}$$

$$= \frac{3}{4\pi^2 T}\frac{1}{J_{\mathrm{L}}(R) + J_{\mathrm{H}}(R)}\mathbf{G}^{-1}(\mathbf{p}) + o\left(\frac{1}{T}\right), \tag{31}$$

whenever $\mathbf{G}(\mathbf{p})$ is nonsingular.

Proof: See the Supplementary Material. ∎

Remark 1: Note that the dependence on $\mathbf{p}$ is entirely through the geometry-dependent centered delay gradients $\widetilde{\mathbf{g}}_i = \mathbf{P}\mathbf{g}_i$.

Under (20)–(21), the PSD of $\mathsf{x}_m(\cdot)$ is $S_{\mathsf{x}_m}(f) = S_{\mathsf{s}}(f) + N_0$ for $|f| \le f_{\mathrm{H}}$, hence the rate constraint (8) becomes

$$R = f_{\mathrm{L}} \left[\log_2\!\left(\tfrac{S_{\mathrm{L}}+N_0}{\lambda}\right)\right]^+ + (f_{\mathrm{H}} - f_{\mathrm{L}}) \left[\log_2\!\left(\tfrac{S_{\mathrm{H}}+N_0}{\lambda}\right)\right]^+. \tag{32}$$

Define the *critical rate* (corresponding to $\lambda = S_{\mathrm{H}} + N_0$),

$$R_{\mathrm{crit}} \triangleq f_{\mathrm{L}} \log_2\!\left(\frac{S_{\mathrm{L}} + N_0}{S_{\mathrm{H}} + N_0}\right). \tag{33}$$

Then $\lambda(R)$ is explicit:

- **High-rates** ($R > R_{\mathrm{crit}}$): both bands are active ($\lambda < S_{\mathrm{H}} + N_0$) and

$$\lambda(R) = 2^{-\frac{R}{f_{\mathrm{H}}}} (S_{\mathrm{L}} + N_0)^{\frac{f_{\mathrm{L}}}{f_{\mathrm{H}}}} (S_{\mathrm{H}} + N_0)^{\frac{f_{\mathrm{H}}-f_{\mathrm{L}}}{f_{\mathrm{H}}}}. \tag{34}$$

  Thus $B_b(R) = 1 - \lambda(R)/(S_b + N_0)$ for $b \in \{\mathrm{L}, \mathrm{H}\}$, and $w_b(R)$ follows from (24)–(25).
- **Intermediate-rates** ($0 < R \le R_{\mathrm{crit}}$): only $\mathcal{B}_{\mathrm{L}}$ is active ($S_{\mathrm{H}} + N_0 \le \lambda < S_{\mathrm{L}} + N_0$) and

$$\lambda(R) = (S_{\mathrm{L}} + N_0)\, 2^{-R/f_{\mathrm{L}}}, \tag{35}$$

$$B_{\mathrm{L}}(R) = 1 - 2^{-R/f_{\mathrm{L}}}, \quad B_{\mathrm{H}}(R) = 0, \tag{36}$$

  so that $w_{\mathrm{H}}(R) = 0$ and accordingly $J_{\mathrm{H}}(R) = 0$.

Along with (33)–(36), (31) provides an explicit closed-form CRLB parameterized by the physically meaningful quantities $(S_{\mathrm{L}}, S_{\mathrm{H}}, N_0, f_{\mathrm{L}}, f_{\mathrm{H}}, M, R, T, \{\mathbf{p}_m\}, c)$. Specifically, when R drops below R_{crit}, RD-optimal compression for signal reconstruction eliminates $\mathcal{B}_{\mathrm{H}}$ (hence $w_{\mathrm{H}}(R) = 0$), even though the Fisher integrand scales as f^2, indicating that higher-frequency components can carry more localization-related information. This mechanism is generally localization-suboptimal and can lead to a catastrophic degradation in localization accuracy.

To see this more clearly, consider the following example. Fix a rate $R \le R_{\mathrm{crit}}$, for which RD-optimal compression satisfies $B_{\mathrm{H}}(R) = 0$ and hence $J_{\mathrm{H}}(R) = 0$. Now, consider instead the following (RD-suboptimal) *band-selective* compression scheme: each sensor suppresses $\mathcal{B}_{\mathrm{L}}$ and applies the Gaussian test channel only on $\mathcal{B}_{\mathrm{H}}$, i.e.,

$$B^{\mathrm{sel}}(f) = \begin{cases} 0, & f \in \mathcal{B}_{\mathrm{L}}, \\ \left[1 - \frac{\lambda_{\mathrm{sel}}}{S_{\mathrm{H}}+N_0}\right]^+, & f \in \mathcal{B}_{\mathrm{H}}, \end{cases} \tag{37}$$

such that $S_{\mathsf{z}}^{\mathrm{sel}}(f) = \lambda_{\mathrm{sel}} B^{\mathrm{sel}}(f)$ on $\mathcal{B}_{\mathrm{H}}$ and zero otherwise. Imposing the *same rate* R over bandwidth $(f_{\mathrm{H}} - f_{\mathrm{L}})$ gives

$$R = (f_{\mathrm{H}} - f_{\mathrm{L}}) \left[\log_2\!\left(\frac{S_{\mathrm{H}} + N_0}{\lambda_{\mathrm{sel}}}\right)\right]^+, \tag{38}$$

$$\lambda_{\mathrm{sel}}(R) = (S_{\mathrm{H}} + N_0)\, 2^{-R/(f_{\mathrm{H}}-f_{\mathrm{L}})}. \tag{39}$$

Hence on $\mathcal{B}_{\mathrm{H}}$,

$$B_{\mathrm{H}}^{\mathrm{sel}}(R) = 1 - 2^{-R/(f_{\mathrm{H}}-f_{\mathrm{L}})}, \qquad B_{\mathrm{L}}^{\mathrm{sel}}(R) = 0. \tag{40}$$

Define $\gamma_{\mathrm{H}}^{\mathrm{sel}}(R)$ and $w_{\mathrm{H}}^{\mathrm{sel}}(R)$ by the same formulas as in (24)–(25), but with (B_b, λ) replaced by $(B_b^{\mathrm{sel}}, \lambda_{\mathrm{sel}})$. Then, the FI rate under the band-selective scheme satisfies

$$\mathbf{J}_{\infty}^{\mathrm{sel}}(\mathbf{p}) = \frac{4\pi^2}{3} J_{\mathrm{H}}^{\mathrm{sel}}(R)\, \mathbf{G}(\mathbf{p}), \quad J_{\mathrm{H}}^{\mathrm{sel}}(R) \triangleq (f_{\mathrm{H}}^3 - f_{\mathrm{L}}^3)\, w_{\mathrm{H}}^{\mathrm{sel}}(R), \tag{41}$$

while under RD (for $R \le R_{\mathrm{crit}}$) we have $J_{\mathrm{H}}^{\mathrm{RD}}(R) = 0$, hence

$$\mathbf{J}_{\infty}^{\mathrm{RD}}(\mathbf{p}) = \frac{4\pi^2}{3} J_{\mathrm{L}}^{\mathrm{RD}}(R)\, \mathbf{G}(\mathbf{p}), \quad J_{\mathrm{L}}^{\mathrm{RD}}(R) = f_{\mathrm{L}}^3\, w_{\mathrm{L}}(R). \tag{42}$$

If $J_{\mathrm{H}}^{\mathrm{sel}}(R) > J_{\mathrm{L}}^{\mathrm{RD}}(R)$, then $\mathbf{J}_{\infty}^{\mathrm{sel}}(\mathbf{p}) \succ \mathbf{J}_{\infty}^{\mathrm{RD}}(\mathbf{p})$ and, consequently,

$$\frac{\left[\mathbf{J}_T^{-1}(\mathbf{p})\right]_{ii}^{\mathrm{sel}}}{\left[\mathbf{J}_T^{-1}(\mathbf{p})\right]_{ii}^{\mathrm{RD}}} = \frac{J_{\mathrm{L}}^{\mathrm{RD}}(R)}{J_{\mathrm{H}}^{\mathrm{sel}}(R)} < 1, \quad 1 \le i \le d, \tag{43}$$

i.e., the CRLB for the band-selective scheme is strictly smaller.

As a more concrete example, consider a mmWave/ISAC-like wideband regime [23], [24] with $M = 4$ sensors, wherein $f_{\mathrm{H}} = 200\,\mathrm{MHz}$, and further fix $f_{\mathrm{L}} = 5\,\mathrm{MHz}$, $S_{\mathrm{L}} = 100$, $S_{\mathrm{H}} = 20$, and $N_0 = 1$. For these values, $R_{\mathrm{crit}} \approx 11.3\,\mathrm{Mb/s}$. For a rate $R = 10\,\mathrm{Mb/s} < R_{\mathrm{crit}}$ (a plausible per-sensor backhaul budget), using the expressions from Theorem 1, we obtain in this setting

$$\frac{\left[\mathbf{J}_T^{-1}(\mathbf{p})\right]_{ii}^{\mathrm{sel}}}{\left[\mathbf{J}_T^{-1}(\mathbf{p})\right]_{ii}^{\mathrm{RD}}} = \frac{J_{\mathrm{L}}^{\mathrm{RD}}(R)}{J_{\mathrm{H}}^{\mathrm{sel}}(R)} \approx 9.97 \times 10^{-3}, \quad 1 \le i \le d, \tag{44}$$

i.e., at the *same* rate R the band-selective scheme yields a reduction of two orders of magnitude in the CRLB.

V. Discussion and Outlook

This letter derived fundamental localization limits when a FC has access only to rate-constrained (compressed) versions of wideband multi-sensor observations. Leveraging a Gaussian line-of-sight waveform model and the Gaussian RD test-channel representation, we obtained an explicit frequency-domain characterization of the Fisher information and the associated CRLB. Specializing further to a simple, yet insightful two-band, two-level spectrum model yielded closed-form expressions that transparently expose the interaction between rate allocation, spectral content, and localization accuracy.

Beyond providing an analytically tractable CRLB, the two-band specialization highlights a key conceptual message: under the Gaussian per-sensor RD benchmark, compression optimized for waveform reconstruction can be poorly aligned with localization. In particular, under RD-optimal compression, decreasing the rate can induce a sharp regime change in which high-frequency components are eliminated when the compression rate is sufficiently low, despite the fact that localization information is inherently weighted towards higher frequencies due to the phase sensitivity. This reveals an explicit mechanism by which compress-then-estimate designs based on reconstruction-optimal per-sensor RD compression may incur a dramatic loss in localization accuracy. This observation, in turn, motivates task- or goal-oriented compression strategies that preserve the most localization-informative signal features.

Naturally, there are many related important questions that remain to be addressed. While our analysis focused on RD-optimal compression under the quadratic distortion criterion, a central direction is to characterize and design *localization-aware* compression rules, e.g., by formulating rate-allocation problems that maximize localization fidelity subject to rate constraints. It is also of both theoretical and practical interest to extend the framework beyond line-of-sight to multi-path and cluttered environments, and to integrate the resulting bounds into ISAC-oriented system design, including bandwidth allocation, and joint sensing–communication resource management.

References

[1] L. D. Xu, W. He, and S. Li, "Internet of Things in industries: A survey," *IEEE Trans. Ind. Informat.*, vol. 10, no. 4, pp. 2233–2243, Nov. 2014.

[2] D. Wu and J. Liebeherr, "A low-cost low-power LoRa mesh network for large-scale environmental sensing," *IEEE Internet Things J.*, vol. 10, no. 19, pp. 16700–16714, Oct. 2023.
[3] T. Sztyler and H. Stuckenschmidt, "On-body localization of wearable devices: An investigation of position-aware activity recognition," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun*, 2016, pp. 1–9.
[4] Y. Lu, D. Wang, M. S. Obaidat, and P. Vijayakumar, "Edge-assisted intelligent device authentication in cyber–physical systems," *IEEE Internet Things J.*, vol. 10, no. 4, pp. 3057–3070, Feb. 2023.
[5] K. L. Ketshabetswe, A. M. Zungeru, B. Mtengi, C. K. Lebekwe, and S. R. S. Prabaharan, "Data compression algorithms for wireless sensor networks: A review and comparison," *IEEE Access*, vol. 9, pp. 136872–136891, 2021.
[6] N. Shlezinger, Y. C. Eldar, and M. R. D. Rodrigues, "Hardware-limited task-based quantization," *IEEE Trans. Signal Process.*, vol. 67, no. 20, pp. 5223–5238, Oct. 2019.
[7] Z. Zhang and T. Berger, "Estimation via compressed information," *IEEE Trans. Inf. Theory*, vol. 34, no. 2, pp. 198–211, Mar. 1988.
[8] S. E. Trevlakis et al., "Localization as a key enabler of 6G wireless systems: A comprehensive survey and an outlook," *IEEE Open J. Intell. Commun. Soc.*, vol. 4, pp. 2733–2801, 2023.
[9] F. Liu et al., "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, Jun. 2022.
[10] A. Liu et al., "A survey on fundamental limits of integrated sensing and communication," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 994–1034, 2nd Quart. 2022.
[11] S. Lu et al., "Integrated sensing and communications: Recent advances and ten open challenges," *IEEE Internet Things J.*, vol. 11, no. 11, pp. 19094–19120, Jun. 2024.
[12] T. Berger, Z. Zhang, and H. Viswanathan, "The CEO problem [multiterminal source coding]," *IEEE Trans. Inf. Theory*, vol. 42, no. 3, pp. 887–902, May 1996.
[13] Y. Shen and M. Z. Win, "Fundamental limits of wideband localization—Part I: A general framework," *IEEE Trans. Inf. Theory*, vol. 56, no. 10, pp. 4956–4980, Oct. 2010.
[14] Y. Shen, H. Wymeersch, and M. Z. Win, "Fundamental limits of wideband localization—Part II: Cooperative networks," *IEEE Trans. Inf. Theory*, vol. 56, no. 10, pp. 4981–5000, Oct. 2010.
[15] G. O. Balkan and S. Gezici, "CRLB based optimal noise enhanced parameter estimation using quantized observations," *IEEE Signal Process. Lett.*, vol. 17, no. 5, pp. 477–480, May 2010.
[16] H. Wymeersch, J. Lien, and M. Z. Win, "Cooperative localization in wireless networks," *Proc. IEEE*, vol. 97, no. 2, pp. 427–450, Feb. 2009.
[17] S. Gezici, H. Celebi, H. V. Poor, and H. Arslan, "Fundamental limits on time delay estimation in dispersed spectrum cognitive radio systems," *IEEE Trans. Wireless Commun.*, vol. 8, no. 1, pp. 78–83, Jan. 2009.
[18] S. Srinivasan, "Cramér–Rao bounds for rate-constrained distributed time-delay estimation," in *Proc. 4th Int. Symp. Commun., Control Signal Process.*, 2010, pp. 1–4.
[19] T. Berger, *Rate Distortion Theory: A Mathematical Basis for Data Compression*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1971.
[20] T. M. Cover and J. A. Thomas, *Elements of Information Theory*. Hoboken, NJ, USA: Wiley, 1999.
[21] P. Whittle, "The analysis of multiple stationary time series," *J. Roy. Statist. Soc. Ser. B*, vol. 15, no. 1, pp. 125–139, 1953.
[22] H. L. Van Trees, *Detection, Estimation, and Modulation Theory, Part I: Detection, Estimation, and Linear Modulation Theory*. Hoboken, NJ, USA: Wiley, 2004.
[23] Z. Gao et al., "Integrated sensing and communication with mmWave massive MIMO: A compressed sampling perspective," *IEEE Trans. Wireless Commun.*, vol. 22, no. 3, pp. 1745–1762, Mar. 2023.
[24] H. Guo et al., "Integrated communication, localization, and sensing in 6G D-MIMO networks," *IEEE Wireless Commun.*, vol. 32, no. 2, pp. 214–221, Apr. 2025.